

PRIVATE B2B INVOICING ON MIDNIGHT

Inkan

Invoices settle on-chain.
Their terms never do.

Japanese consumption-tax rules checked inside the proof · shielded settlement · selective audit

THE PROBLEM

On a public ledger, an invoice tells everyone everything.

COMPETITORS
read your prices.

SUPPLIERS
read who your customers are.

LENDERS
read how late you pay.

So businesses invoice off-chain, where nothing is provable. **Privacy alone does not fix it:** books get audited, taxes get filed, and Japan's qualified-invoice system requires the tax to be correct for each rate. The terms must stay private and the proof must stay public.

WHAT INKAN DOES

Four operations, one contract.

CREATE

The seller's terms stay in their private state. The chain records commitments, and the contract refuses tax groups that do not hold.

PAY

The buyer pays in shielded NIGHT inside the transaction that marks the invoice paid. The contract forwards the coin; it never holds it.

CANCEL

The seller can cancel a pending invoice. A paid invoice cannot be retracted.

AUDIT

The seller lets one auditor see chosen fields of one invoice until a date, and nothing else.

WHAT THE CHAIN SEES

Hidden by default. Provable on demand.

PUBLIC, PER INVOICE

Invoice identifier
Two pseudonymous party keys
Status: pending, paid, cancelled
Commitment to the full terms
Eleven per-field commitments and their root
Salted tax tag, rule-result digest
Audit grant: key hash, field set, expiry

PRIVATE, HELD BY THE PARTIES

Amount, tax and total
The two tax-rate groups
Due date and currency
Payout addresses
Line-item and memo digests, order reference
Every commitment opening

Terms reach a circuit only as witnesses. Tests read the ledger back and fail if any private figure appears in it.

TAX RULES INSIDE THE PROOF

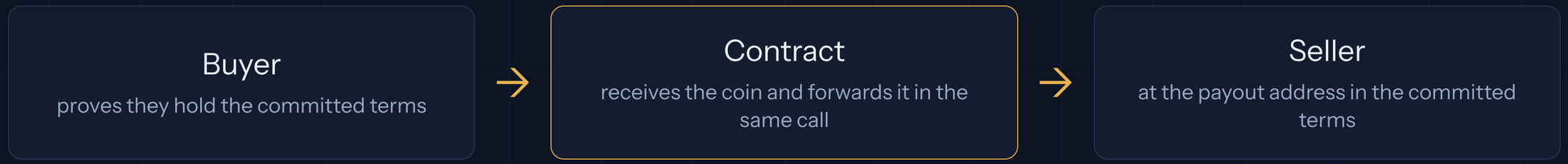
The contract refuses an invoice whose tax is wrong.

```
// for each rate group of a qualified invoice (10% standard, 8% reduced)
assert tax * 10000 == net * rate        // rate in basis points
assert (net10 + tax10) + (net8 + tax8) == total
```

Compact has no division, so the rule is stated as a product. It is exact for every amount: 15,000 at 10% is 1,500, where dividing first, (net / 10000) × rate, gives 1,000. The verifier learns that the groups are consistent and add up, and nothing about the figures.

SHIELDED SETTLEMENT

Payment and status change are one transaction.



- The coin must equal the invoice total, in the invoice's token.
- It must go to the seller the invoice names; a prover that lies about the terms is refused.
- Paying after the private due date is refused, against the block's clock.

Next step Tested in process; on Preview it needs **shielded** NIGHT, and the faucet pays unshielded.

SELECTIVE AUDIT

Show one auditor the chosen fields, and nothing else.

SCOPE

Nine fields to choose from: amount, tax, due date, buyer and more. Stored as a nine-slot vector, because Compact has no bitwise operators.

EXPIRY

The grant ends at a date checked against the block, not against a time the caller chose.

KEY

Only a hash of the audit key reaches the chain.

One salt per field. If every field shared one salt, disclosing one would help an auditor test guesses against the rest. With a derived salt per field, it does not.

LIVE ON MIDNIGHT PREVIEW

Four transactions on a public network.

Contract `15efb99ca21bb255b0f12a1de56c0519d3f98e95c4e1e4f341aea8fa0f4ec2a8`

STEP	TRANSACTION	BLOCK	INDEXER
Deploy	`bbaa1cd4869986b5…41af00b3`	1,364,749	success
createInvoice	`b3349368d698aebb…e262eebe`	1,365,672	success
setAuditAuthorization	`d3d77d267860fb50…87e642c5`	1,365,677	success
cancelInvoice	`e322ff0993f8198c…88ec9374`	1,365,704	success

After each transaction the state was read back through the public indexer and searched for the invoice's amount, tax, total and due date; none appeared. The four-circuit deploy used **23.6%** of the block's write budget. Hashes link to the explorer; all of them are in **docs/receipts/preview.json**.

HOW IT WAS VERIFIED

Tests that were made to fail first.

88

simulator tests against the compiled contract, all passing

audit 12 · create 38 · identity 4 · settle 34

45 / 45

deliberately broken rules caught by the suite; 1 more is provably unreachable and named, not counted

4

transactions on Preview, each read back from the public indexer

The tests cover who may act, wrong amount, wrong token, wrong payee, a prover lying about the terms, the time bounds, the tax rules, and that no amount or payout address reaches public state. **The mutation sweep found a gap the green suite hid**: nothing checked that party keys depend on the deployment salt. Four tests closed it.

ARCHITECTURE

Private state on the device. Commitments on the chain.

Interface	Next.js 14, English, Japanese and Chinese. Not yet connected to the Midnight contract; next step, through the Lace wallet.
Private state & witnesses	The seller's secret and the open terms, handed to circuits as witnesses with size and non-zero checks (witnesses.ts)
Compact contract	Compact 0.23, toolchain 0.31.1: createInvoice, payInvoiceShielded, cancelInvoice, setAuditAuthorization; hashes exported as pure circuits so the app computes exactly what the chain does
Proving & network	Local proof server 8.1.0, Midnight.js 4.1.1, wallet SDK 1.2.0, Zswap shielded coins, DUST fees, the public Preview indexer

Identity is a key derived from a secret the caller proves they hold, never **ownPublicKey()**. Every claimed time is bounded to within one hour of the block and never ahead of it.

WHO IT IS FOR

Businesses that must prove their books without publishing them.

SELLERS

Invoice and get paid on-chain without showing prices or customers to anyone reading the ledger.

BUYERS

Pay in the same step that records the payment, with a settlement anchor they can later show an auditor.

AUDITORS

Check exactly the fields they are shown against commitments the chain has held since the invoice was created.

Why Japan first: the qualified-invoice system makes per-rate tax correctness a legal requirement, and that is a rule a proof can check. Later, credit proofs from settled invoices let a seller show a lender a payment record without the invoices behind it. Pricing is not set yet.

WHERE IT STANDS

Built, tested and deployed; next, connected.

DONE

- Contract with four entry points, full proving keys
- 88 tests, 45 of 45 mutants caught
- Deploy, create, audit grant and cancel on Preview
- Demo video, receipts, Apache-2.0

NEXT

- The payment on Preview, with shielded NIGHT
- The interface on Midnight, through Lace
- Escrow and disputes, in the same contract
- Credit proofs from settled invoices

Code: github.com/Ritik200238/inkan · video: docs/inkan-demo.mp4 · receipts: docs/receipts/preview.json